# Macintosh PiForma
# Owner's Guide



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# Welcome to the Macintosh PiForma

Welcome to the Macintosh PiForma.

This little computer began in the basement of Apple headquarters, which is remarkable, because Apple headquarters did not have a basement. A rogue engineer came across an Apple IIc Monitor down there, emptied it, and Frankensteined a computer into the case from whatever parts were lying around, several of which had not, strictly speaking, been invented yet.

The work went on under the nose of the vice president of the day, a man with firm opinions about computers that customers could open. The engineer was in no hurry to show him a Macintosh with a latch, running open-source software and other people's good ideas.

Then, in 1985, the vice president was let go, and the mood changed. The CEO turned out to be rather more open-minded about unconventional ideas than anyone had expected. The project came up out of the basement, and the PiForma was given the green light. Management has learned to live with the latch.

Only one PiForma was ever made, and this is it. Its story is almost entirely unknown; you may be the first person ever to read it.

Inside, the old monitor's case holds a modern display and a battery-powered computer, with a familiar-looking desktop, games, and a few diversions that have earned their place mostly by being amusing. Its name is a small joke; its carrying bag is a surprisingly serious undertaking. You can use it without knowing how the wiring works, although you will want to know which way the keyboard goes back into the bag.

If you have found the PiForma on display, you are welcome to leaf through this guide without touching the computer. It explains what the PiForma does, what is inside it and how it works, laid out the way Apple laid out the manuals that came with its own computers.

This guide starts with the things you need to touch, connect, and switch on. Later chapters explain the parts and programs behind the appearance. There is no required course in Raspberry Pi administration before your first game.

# About this manual

Chapter 1, “Getting Started,” explains how to unpack the computer, arrange the equipment, check the battery, start up, and stop. It also puts everything back into the bag in the right order. Taking things out is easier than remembering where they belonged.

Chapter 2, “Inside the Macintosh PiForma,” introduces the monitor shell, Raspberry Pi, display, battery, controls, keyboard, mouse, speakers, and bag. It explains what the parts do before getting into the finer details.

Chapter 3, “How the Macintosh PiForma Works,” introduces the desktop, launcher, menu bar, games, and other programs. It distinguishes the computer’s physical controls from the menus on its screen.

Chapter 4, “Caring for Your Macintosh PiForma,” covers handling, cleaning, charging, storage, and transport. It includes the brief but comprehensive description of the service department.

Appendix A collects technical information and points to the project documentation. Appendix B helps you find a useful check when something goes wrong. The glossary explains unfamiliar terms used in the guide.

Start with Chapter 1 for your first session. Return to Chapter 3 when you want to work with programs and files, or to Appendix B when the computer has other plans.

CHAPTER 1

# Getting Started

This chapter will help you unpack your Macintosh PiForma, set up its traveling companions, and start the computer. You will also learn how to check a reluctant battery, shut down properly, and return everything to the bag. The packing order is worth a little attention. The foam has opinions.



## Unpacking the equipment

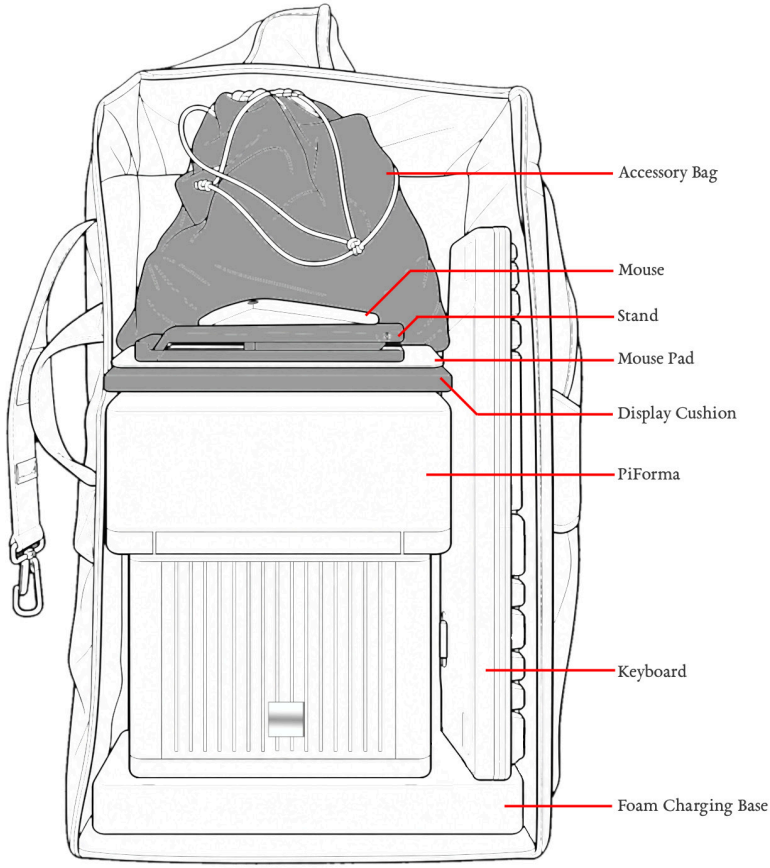
Set the carrying bag upright on a stable surface and open it. Leave room nearby for each item as you take it out.

1. **Remove the small brown accessory pouch near the top.** It contains two Nintendo 64 controllers, the USB hub, and the power cable. Set the pouch and its contents aside.
2. **Take out the mouse and the collapsible computer stand beneath the pouch.**
3. **Remove this printed guide.** If you are reading it, you have already done this step. Well done.
4. **Lift out the mouse pad beneath the guide.**
5. **Remove the velvet-covered foam board beneath the mouse pad.** Keep it flat; it protects the computer when everything is packed.

The PiForma is underneath the board, with its monitor/screen facing upward. The keyboard is tucked against one side of the computer. It is not another layer in the stack. Note which way it faces, as described under “The keyboard’s place” below, and lift the keyboard out first, before the computer.

Lift the PiForma by its handle and case. Do not push on the screen: pressing on the display can damage the screen or its mounts.

## Unpacking the equipment




■ Arrangement in the carrying bag. The printed guide is not shown; it lies beneath the mouse and stand, above the mouse pad.

### Finding the handle

The carrying handle may be hidden against the inside front wall of the bag. Slide a hand down inside that wall, along the top side of the PiForma, until you can pull the handle outward. Do not pull on a connector or an edge of the display to lift the machine.

If lifting it out upright is awkward, lay the bag on its back first and try the gentler removal angle. Lift carefully. There is no prize for wrestling it out upright.

### The keyboard's place

Notice the keyboard's orientation before moving it: the keys face the bag wall, the thicker edge faces the rear of the bag, and the thinner edge faces the front. Remember these three directions when you pack it again.

The molded edges in the bottom foam locate the rear/bottom of the computer. The first photograph shows the empty base; the second shows the PiForma and keyboard in place. Match the keyboard's orientation, rather than merely finding a gap large enough to hold it.



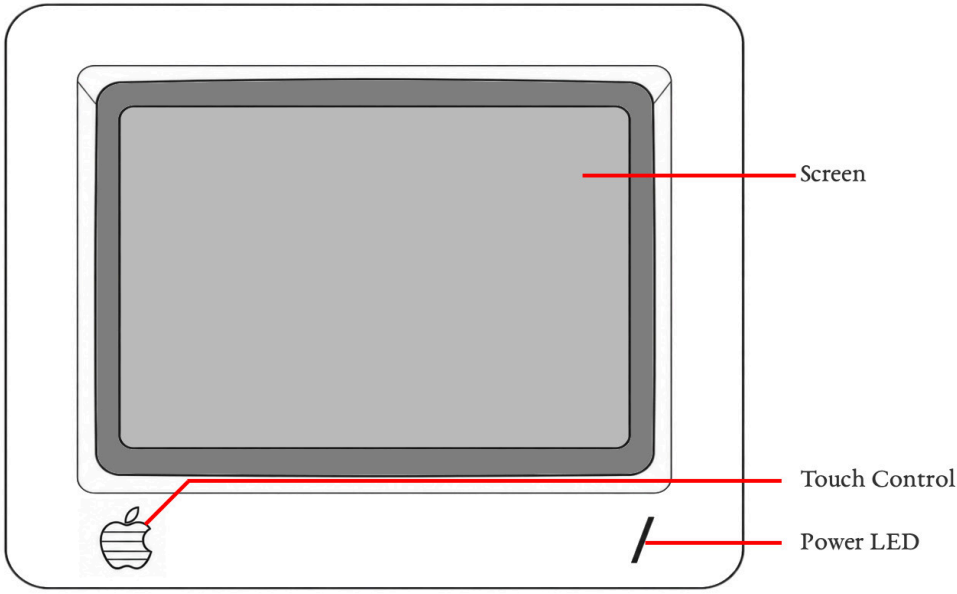
The bottom foam insert. Seat the rear/bottom of the PiForma against the molded edges.



PiForma and keyboard inside the bag. Keys face the bag wall; the thick edge goes toward the rear, the thin edge toward the front.

## Taking a closer look

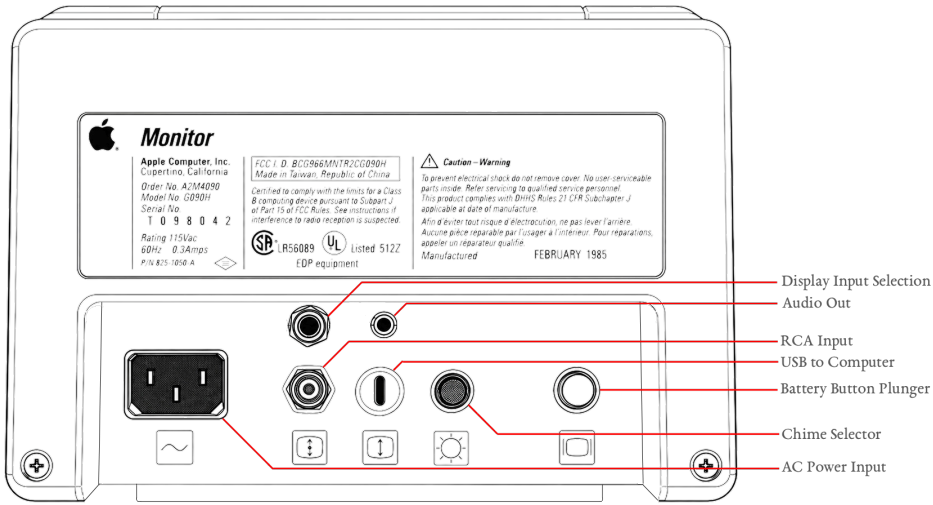
Before connecting anything, find the power control on the right side, the green status light at the front, and the chime-selector dial at the back. The Apple logo at the front is also a touch control. It is separate from the physical power switch.




The front: screen, Apple-logo touch control, and power indicator.

The owner's current arrangement has one externally accessible USB port at the back. The USB hub lets you attach more than one USB device. The keyboard communicates over Bluetooth; the mouse uses a USB receiver. A keyboard connection problem and a missing mouse receiver therefore call for different checks.

At the rear are a display-source selector, a composite-video RCA input and an audio output socket. That RCA jack is for video, not a general audio connection. The audio socket sends sound out to external speakers or headphones. Use the source selector's HDMI setting for the Raspberry Pi picture.

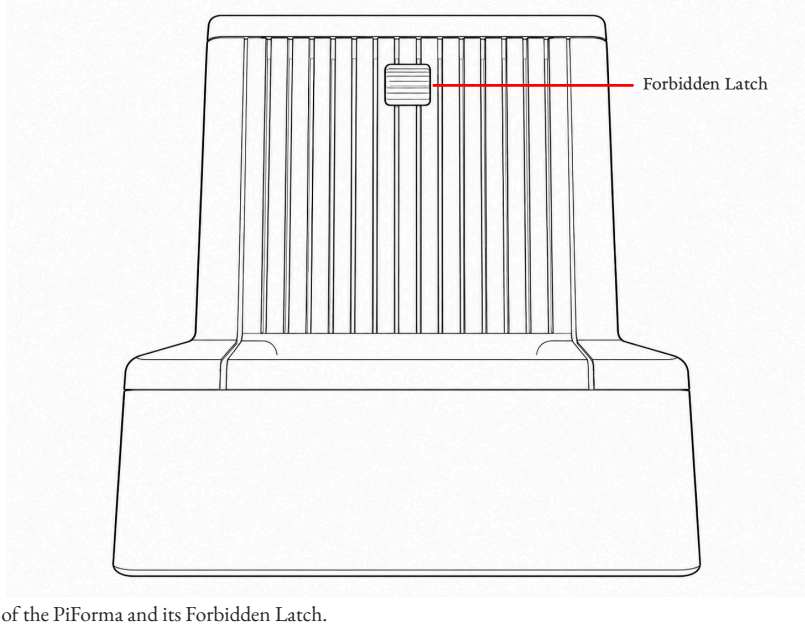



Rear controls and connections. The RCA input is composite video; the audio socket is an output for speakers or headphones.

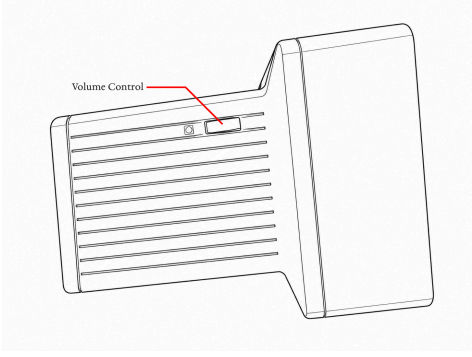
## The forbidden latch

The slider on top releases the upper compartment. Its official designation, at least around here, is the Forbidden Latch. Management would prefer you to admire it without touching it. I have gone rogue and made it conveniently openable. Please pretend I never told you; I would like to keep both my job and my continued existence.

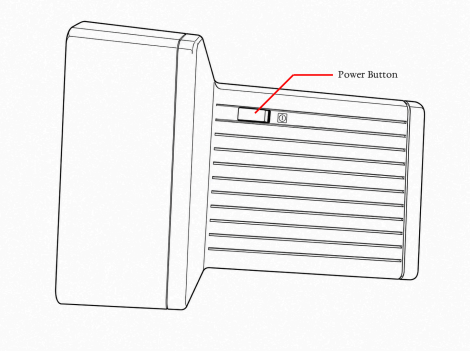
"A closer look, strictly off the record", later in this chapter, explains how it opens. You can check the charge through the top slits without opening it, which should keep management happy.




The top of the PiForma and its Forbidden Latch.




The side volume control.




The physical power button on the opposite side. This is separate from the Apple-logo touch control.

## Basic ergonomics

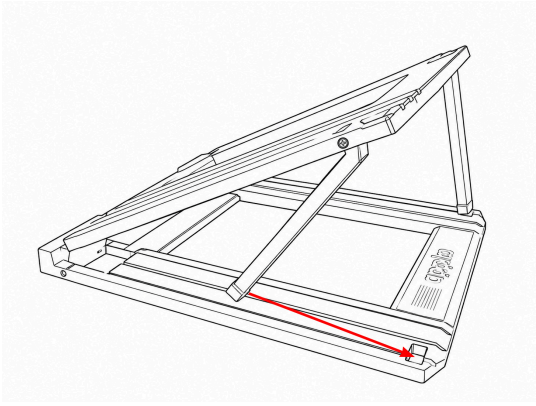
Choose a solid, level table with room for the stand, keyboard, mouse pad, and cables. Place the screen at a comfortable viewing angle, the keyboard directly in front of you, and the mouse beside it. Keep light from reflecting directly off the display. Adjust your chair rather than spending the afternoon leaning toward the computer.

Leave room to move your hands and take occasional breaks. Put speakers and their stands where they will not tip, block the screen, or pull on the computer's cables. A small computer still deserves a sensible place to sit.

## Putting it all together

### The stand, keyboard and mouse

1. **Open the collapsible stand and place it on the level table.** Seat the support leg in its final position, as the arrow in the illustration shows. Check that it is stable before adding the computer.
2. **Place the PiForma on the stand and center it.** Check that it rests securely rather than leaning toward one side.



Seat the support leg in the position indicated by the arrow, then check the stand before adding the computer.

3. **Put the keyboard directly in front.** Turn it on with the switch near its top-right edge.
4. **Put the mouse on its pad beside the keyboard.** Its power switch is underneath; turn it on if necessary.

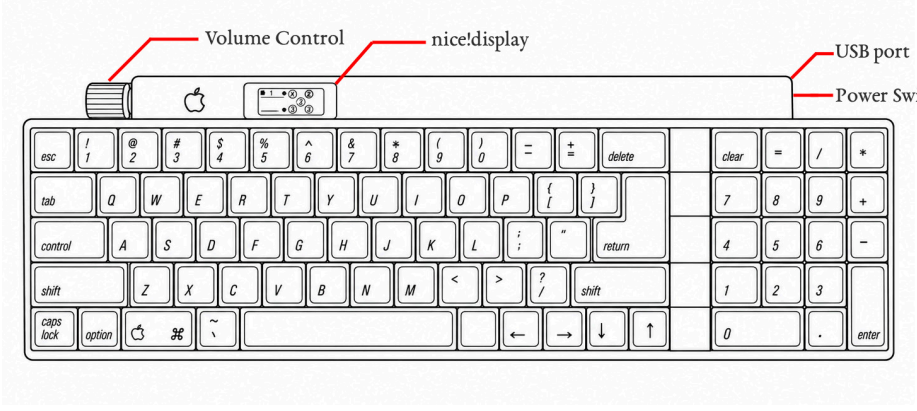
The mouse can ordinarily remain on between uses because it enters a low-power idle state. You can still switch it off for transport or prolonged storage. This does not switch off the keyboard, which has its own control and Bluetooth connection.

The stand illustration deliberately shows one support leg short of its final position. Move it into the seat indicated by the arrow before placing the PiForma on top. The partly assembled position is not the position for supporting the computer.

## The keyboard controls

The overhead view identifies the keyboard's volume control and nice!view status display. The USB charging port and power switch are in the indicated area near the top-right edge; their labels show the general location because the port and switch themselves are not visible from above. The USB-C port sits on the right, where the keyboard's ADB port used to be; charge the keyboard there.

Caps Lock deserves a warning. It is a vintage locking key, and on the PiForma it behaves like a Shift key held down. While it is down, letters come out as capitals and the number keys type symbols. Make sure Caps Lock is up, level with Shift and Option, before you start and before you pack the keyboard away.

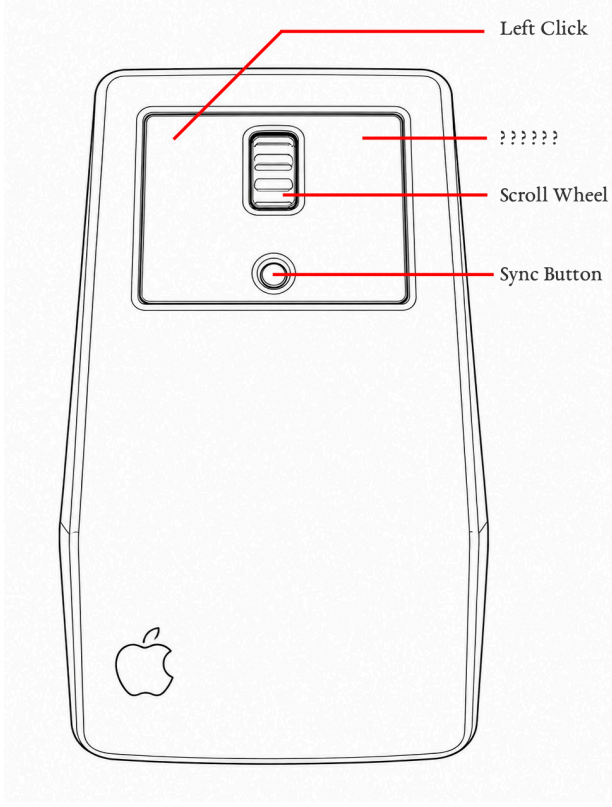



Keyboard controls. The USB port and power-switch callouts indicate their general location, not a directly visible socket or switch.

## The mouse controls

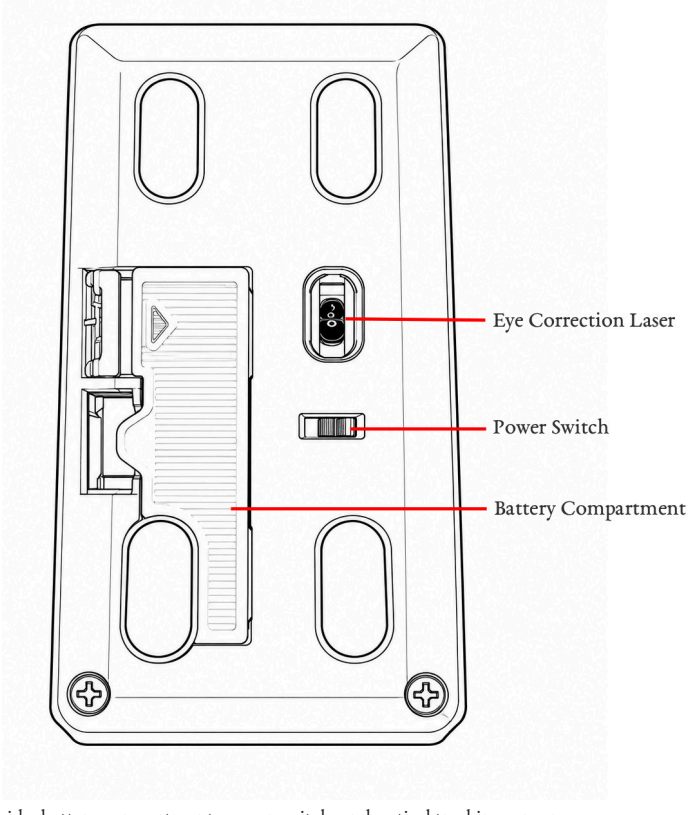
Use the left button for an ordinary click and the wheel for scrolling. The other button is the right button, although the illustration is maintaining a respectable air of mystery. Perhaps it is a second left click. We have never heard of a right click.

The Sync button is not needed in the normal setup: this mouse is used with its one receiver. Leave it alone for ordinary use.




The mouse buttons, scroll wheel, and Sync button. The second button is awaiting official recognition.

Underneath are the battery compartment, power switch, and optical tracking window. The mouse runs on one AA battery; it is not recharged. The illustration calls that last item the Eye Correction Laser. Its job is to follow the mouse pad, not improve your eyesight. Keep it pointed at the desk; no eye treatments are included.




Mouse underside: battery compartment, power switch, and optical tracking sensor.

**Power for a longer session**

If the internal battery is charged, you do not need a power cord for a short session. A full charge typically lasts about six hours of general use, or about four hours of emulated games, Nintendo 64 games especially. For longer sessions, keep the PiForma connected to external power when it is available.

The desktop has no battery gauge, and there is no low-battery warning. Check the battery display through the top slits now and then. When the battery runs out, the PiForma stops at once, just as if the power switch had been turned off, and unsaved work is lost. After a full drain, connect external power and allow a minute or two before it will start.

Take the power cable from the brown pouch. The original rear AC inlet supplies the internal charger. The battery can charge while the computer's power switch is off. Use the cable and charging arrangement intended for this build.

### USB devices and controllers

Use the rear USB port for the USB hub when you need more than one external USB connection. The two Nintendo 64 controllers are in the accessory pouch. Chapter 3 explains how to approach the game launcher and its controls. Use only the leads or adapters supplied for these controllers; do not force a connector into the USB hub.

### Bose speakers and stands

The Bose speakers are an optional extra, not part of the PiForma package; they do not fit in the bag. The pair shown are 1984 Bose RoomMate speakers. They are almost as large as the PiForma's screen, which makes for a great deal of sound from a very small picture, and they were chosen because their proportions match.

Connect the speakers to the rear audio output socket. Switching sound between the internal speaker and the rear socket is not yet a menu choice; ask the builder if it needs changing. Give the speakers a stable place beside the working area, with cables out of the way and no strain on the PiForma. Their stands should not make the table or the computer's folding stand unstable.




The complete setup: the PiForma on its stand, the keyboard in front, the mouse on its pad, a Bose speaker on its stand at each side and a Nintendo 64 controller at the left.

### Bringing sound into the PiForma

The external audio adapter brings sound from an external source into the PiForma over USB so it can play through the normal speakers. This is external audio into the computer, not the computer's ordinary audio output. Use the adapter intended for this setup. The rear RCA jack carries composite video; it is not an audio input.

## Important safety instructions

Keep liquids away, protect the display, and leave ventilation openings clear. Use a stable stand on a level table. Do not carry the computer by cables or press on the screen to move it.

The power switch does not disconnect external charging power from the internal charger. Switching the computer off alone does not make internal electrical work safe. Do not disassemble it with charging power connected. Internal repairs and battery disconnection belong with someone familiar with this particular build, not in a first-start procedure.

Stop using the computer if it becomes unusually hot, smells hot, or has damaged power wiring or a damaged battery. Do not keep retrying power-on in that condition. Use the intended charging equipment. This privately built device has no claimed electrical-safety certification.

## Starting up the computer

**The startup chime**

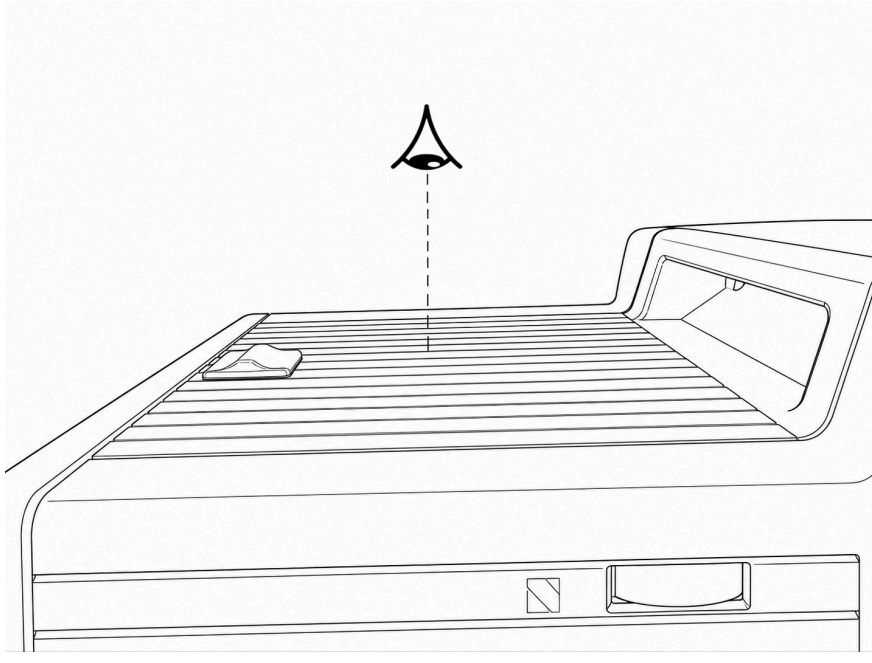
A silent startup does not by itself mean that the computer has failed. Use the green light and display as well as the chime when checking what has happened.

1. **Find the physical power control on the right side and switch the PiForma on.**
2. **Look for the green light at the front.**
3. **Listen for a startup chime.** Depending on the rear selector's position, there may be no sound.
4. **Watch for the startup picture or sequence.** The desktop normally follows in about one to two minutes; allow for some variation.

### Checking the battery

If nothing happens, check the battery before assuming something has broken. The outside plunger mechanically presses the button on the internal battery pack, showing its charge indicators; pressing it may also wake or enable the pack.

The normal, management-approved way to read the battery is to press the rear battery plunger to turn on its display, then peer through the slits in the top to see the percentage. You do not have to open the compartment for this check.



Read the battery percentage through the top slits after pressing the rear battery plunger. The eye shows where to look.

Press the plunger as intended, look at the battery indication, and try the right-side power control again. If the battery is empty, apparently asleep, or refuses to start despite showing charge, connect the intended external power supply and try again.

The owner has sometimes recovered a start after roughly a minute on external power, even when the battery previously showed full. Try a brief connection before taking anything apart, but do not count on one minute as a fixed recovery time. This is an observed oddity, not an established diagnosis.

### If there is no green light

If there is no green light, no chime, and no picture, start with the physical power control, battery indication, plunger, and intended power cable. Retry the startup sequence after the battery check or brief external-power connection. These observations do not yet separate every power fault from every display fault. If they do not help, stop short of internal troubleshooting and consult the builder.

### If the light is on but there is no chime

Some positions on the rear chime-selector dial are intentionally silent or empty. With the PiForma powered on, move the dial through several positions in either direction and listen for an available sound, as the owner suggests.

If it never sounds, there may be a problem with the chime module, its speaker, or its power, but silence alone is not a diagnosis. The startup chime uses its own sound hardware; normal desktop audio is a separate path. Internal checks should be made only by someone familiar with the build and with power safely disconnected.

### If the light and chime work but the screen stays blank

Allow time for the startup picture. Check that the display source is set for the Raspberry Pi's HDMI connection. If the screen remains blank, the display side of the system needs attention; a chime does not prove that the Pi has completed its boot.

Do not disturb internal connections while the machine is powered; ask the builder to investigate a persistent blank or distorted picture. Appendix B provides other checks by symptom.

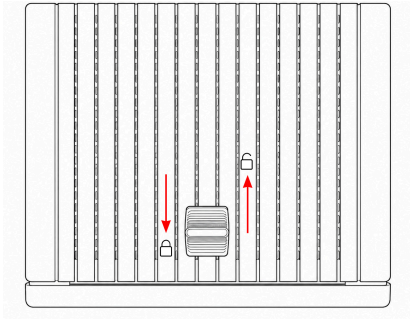
### If the desktop is slow to appear

Give the computer approximately one to two minutes to reach the desktop. Boot time can vary. If it remains at the startup picture or never reaches a usable desktop, record what you see rather than repeatedly interrupting power.

**A closer look, strictly off the record**

For an optional open-top look, first shut down the PiForma, switch off physical power, and disconnect external power. Opening the compartment is for looking, not changing internal wiring.

1. **Slide the top latch toward the open-padlock mark: upward, as the illustration shows.**
2. **With the latch unlocked, lift at the slider to open the upper compartment.**
3. **Close the compartment before sliding the latch toward the closed-padlock mark, downward in the illustration, to lock it again.**



Slide up toward the open padlock to unlock; lift at the slider to open. Close the top before sliding down to lock.

Of course, you did not hear any of this from me. If anyone from management asks, you were merely admiring the ventilation slits. I would appreciate remaining employed. And alive.

## Your first session

Once the desktop appears, try this short tour. You need not change any settings or make a file.

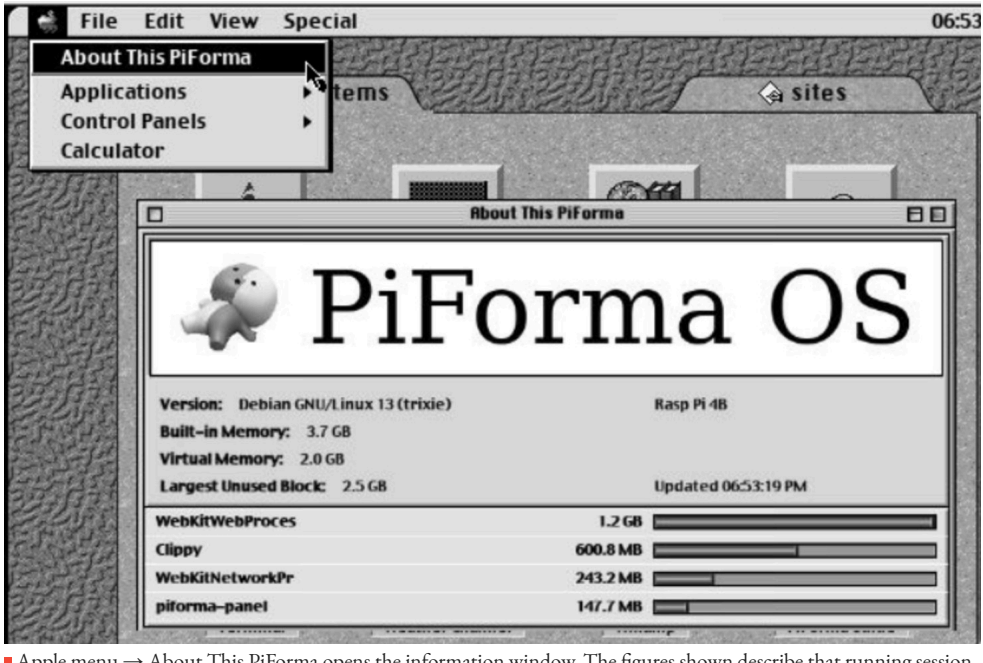



■ The PiForma desktop: persistent system menu bar across the top, AtEase in the center, Control Strip at lower left, and Clippy at lower right.

1. **Move the mouse on its pad and watch the pointer follow it.** Point at an item by moving the pointer over it. To click, press and release the left mouse button once. The other left button can wait for Chapter 3.
2. **Find the menu bar at the top of the screen and the folder-like AtEase launcher beneath it.** If a window covers the launcher, choose View → Show Desktop from the top menu bar. This reveals the desktop without quitting your programs.
3. **In AtEase, click a folder tab if needed, then click a program button once.** Wait for its window to appear before clicking again. Use a windowed program for this first exercise; a full-screen game will cover the desktop.

4. **Explore the program's visible controls.** When finished, use its own Quit or Exit command if it has one, or its window close control. If asked to save something you want to keep, save it before continuing. Closing one window may leave other windows from the same program open.
5. **Choose View → Show Desktop to return to the launcher.** You can now open another program in the same way.

For a small practice window with a known menu entry, choose Apple menu → About This PiForma. Read the information, close that window using its close control, and return to AtEase. Opening a program, leaving it, and finding your way back are most of the battle. Chapter 3 explains windows, typing, files and games in more detail.



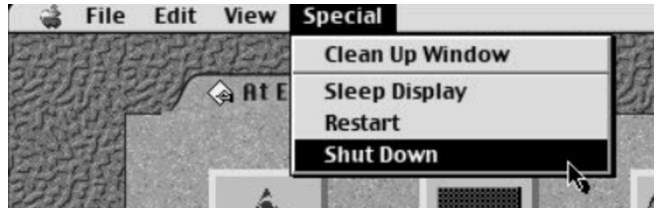

Apple menu → About This PiForma opens the information window. The figures shown describe that running session, rather than fixed memory-use specifications.

The About window is not the project guide. To read the guide itself, click PiForma Guide in AtEase; Chapter 3 describes it.

If the launcher or menu bar is missing, see Appendix B rather than trying unfamiliar touch gestures. When you have finished using the computer, follow "Stopping the computer" below.

## Stopping the computer

1. **Save your work and leave any game or application using its own quit command when available.**
2. **In the PiForma Panel menu bar, choose Special → Shut Down.**
3. **If a confirmation appears, choose Shut Down.** Let the operating system finish shutting down before removing physical power. A blank screen alone can also mean the display is asleep, so do not use darkness as the only confirmation.
4. **Once shutdown is complete, turn off the right-side physical power control.** The wiring permits battery charging while this control is off.




Choose Special → Shut Down after saving your work. This view shows the menu choice, not completed shutdown.

The physical power switch can cut the computer's power. Reserve an abrupt power-off for a last resort when the system will not respond, rather than treating it as the normal way to stop. It can lose unsaved work or damage files.

To leave a full-screen program rather than stop the computer, see "Leaving a full-screen program" in Chapter 3. A hold of about six seconds closes only the active application on the current PiForma.

## Packing the equipment

1. **Shut down properly, switch off physical power, and disconnect external cables as needed.** Make sure the computer and accessories are cool and dry before putting them away.

2. **Place the PiForma in the bag with its monitor/screen facing upward.** Its rear/bottom must seat in the molded indents in the bottom foam insert.
3. **Gently shift or wiggle it into its intended resting place.** It should not move much when seated. Do not force it down or put pressure on the display.
4. **Put the keyboard beside the computer: keys toward the bag wall, thicker edge toward the rear, thinner edge toward the front.**
5. **Place the velvet-covered foam board above the computer.**
6. **Put the mouse pad above the board, then this guide above the pad.**
7. **Put the mouse and collapsible stand above the guide.**
8. **Return the two controllers, hub, and power cable to the brown pouch and place that pouch at the top.** Check that no loose item presses against the screen and close the bag gently.

### Checking the charging alignment

Seating the rear/bottom in the molded foam indents aligns the bag's wireless-charging arrangement. A hidden transmitter sits under the velvet, with its USB power cable routed into a back pocket. This is a slow, trickle-charging arrangement, not fast charging.

Physical fit does not prove that charging has started. Check the foam seating and that the charging cable in the back pocket has power. The battery display can indicate whether power from the wireless receiver is charging the battery, but the display is hidden while the computer is packed in the bag. Without a USB power monitor on the bag charger's supply, it is rather a hope-and-pray arrangement.

A USB power monitor in the bag charger's USB supply connection can show whether it is drawing power. That gives you more evidence than a hopeful glance at the bag, but it is not a direct measurement of energy reaching the battery. Do not press the computer farther into the bag in search of a better fit.

**Carrying the bag**

Lift and carry the closed bag carefully. The padding offers useful protection against ordinary handling; it does not make the computer immune to hard impacts. Cushioning is not armor. The Bose speakers and stands do not fit in this bag; carry them separately.

CHAPTER 2

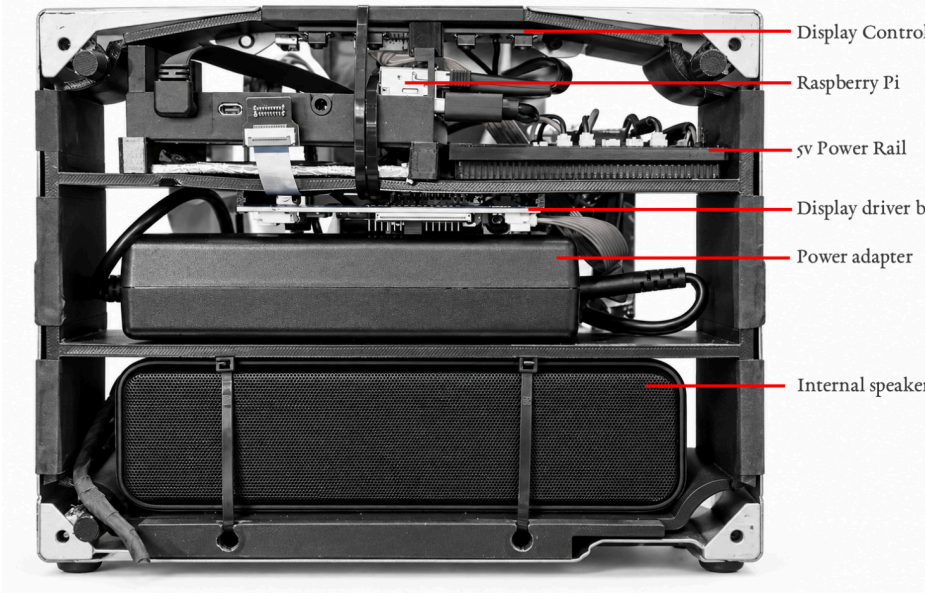
# Inside the Macintosh PiForma

The PiForma's familiar case is the starting point, not the specification of the computer inside it. This chapter introduces the parts and their jobs. Appendix A keeps the finer details close at hand without requiring you to read a wiring inventory before switching on.

## The monitor that became a computer

The case began as an Apple IIc Monitor. Its original CRT had been removed before this build. The plastic case, metal chassis, power switch, AC inlet, and rear openings were reused; the computer and display inside are different equipment.

This is a conversion, not a restoration of a stock Macintosh. The old case was a good place to start and a rather particular place to fit everything else.




The internal assembly, with the Raspberry Pi, display hardware, power adapter, 5V power rail and ordinary internal speaker identified. This is a component tour, not a disassembly procedure.

## Raspberry Pi, memory and storage

A Raspberry Pi runs the programs. RAM holds the work the computer is doing now; the microSD storage card keeps the operating system, applications, and saved files when power is off. Saving transfers your work into a file that can remain after shutdown.

The computer runs Raspberry Pi OS beneath its Macintosh-inspired appearance. Appendix A lists the hardware and software specifications.

## The display and source selector

An LCD screen replaces the original monitor display. The Raspberry Pi supplies its picture over HDMI. A rear RCA input supplies composite video from an external source: a VCR, an older games console or anything else with a composite video output. Its sound comes in separately, through the external audio adapter.

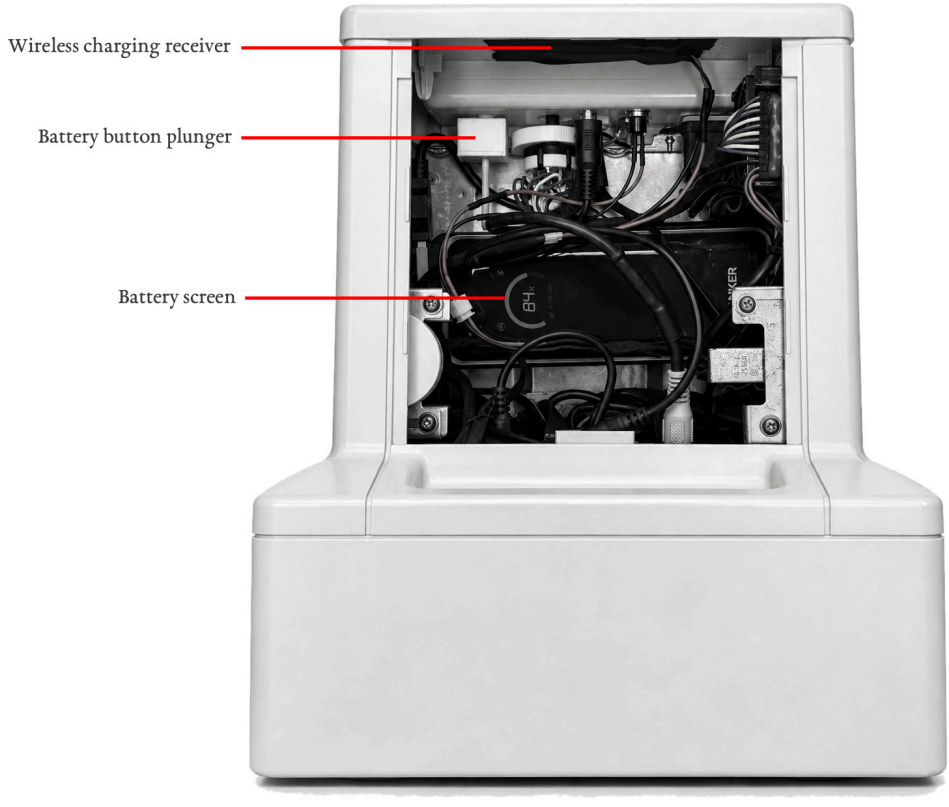
The screen in this first revision is, frankly, of modest quality: a little dim, with colors that are not quite right, and neither brightness nor color can be adjusted from the computer. We apologize for the economy. You make do with what you have.

The rear physical source control cycles through HDMI, RCA, VGA, and RCA2 in the documented build. HDMI and RCA are the normally useful inputs; the existence of other controller modes does not mean that matching external sockets have been fitted. Select HDMI to return to the Pi's picture.

## The battery and charging paths

The internal battery lets you use the PiForma without a power cord. The intended external power cable supplies its charger, and the battery can charge while the computer's physical switch is off. The switch does not disconnect the charger from external power. Appendix A describes the internal power path.

An external plunger reaches the battery's own button. The owner uses it to check charge and sometimes wake the pack. The normal check is to press the rear plunger and read the percentage through the top slits. The owner has also confirmed that the top latch can be unlocked and lifted for a closer look; Chapter 1 describes the optional opening procedure.




■ The open top reveals the wireless charging receiver, battery-button plunger and battery display. The battery is at 84% in this photograph, as confirmed by the owner. Its display can also indicate charging from the wireless receiver, but is hidden while the computer is packed in the bag. For the normal check, read through the top slits as described in Chapter 1.

## System sound and startup chimes

The startup chime and ordinary system audio are separate. Ordinary sound plays through the internal speaker behind the screen or, when selected, through the rear audio output socket. A separate sound module and speaker play the startup chime; the rear selector chooses among its sounds. Some positions are silent or empty.

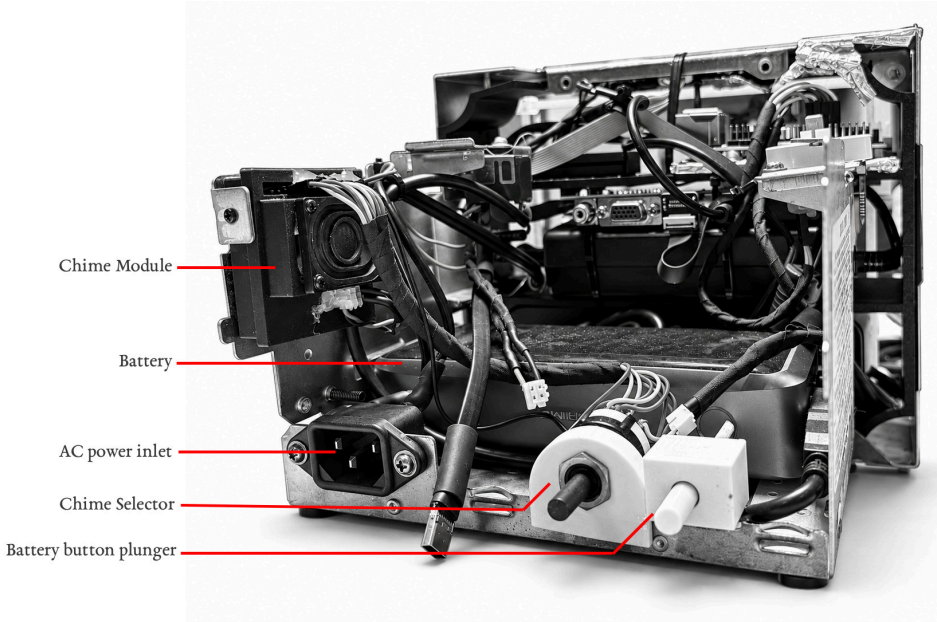
### Startup chimes

Turning the selector one click at a time gives these sounds, in order:

1. The classic Macintosh startup chime.
2. An older Macintosh chime.
3. A Macintosh Plus floppy drive working at startup.
4. The Game Boy Advance startup sound.
5. The Apple II beep.
6. The PlayStation 2 startup sound.
7. A picture tube warming up.
8. The Windows 98 startup sound.
9. The two-note Windows 3.1 fanfare.

The next three clicks are silent, and the one after that brings back the classic chime.

Normal desktop sound uses an internal USB speaker. Changing its software volume is not necessarily a test of the separate startup-chime hardware.




■ A rear oblique view identifies the separate chime assembly, battery, AC power inlet, chime selector and battery-button plunger. The ordinary internal speaker appears in the first photograph of the interior.

## The physical controls

The right-side power switch controls the computer's power rail. The front Apple-logo sensor supplies software gestures. The rear source selector chooses the LCD input, while the rear chime dial chooses the startup sound. They do different jobs even when the desktop is not yet visible.

Turn the side volume knob clockwise to raise normal system volume and counterclockwise to lower it. The installed knob has no usable push-to-mute action.

## Ports, wireless links and adapters

One rear USB port is available to the user. The Pi board's own connectors should not be mistaken for externally accessible ports. Use the hub for multiple peripherals.

The keyboard uses Bluetooth LE and the mouse uses its USB receiver. Wi-Fi and Bluetooth belong to the operating system's network and device tools. External audio input uses a separate USB capture adapter and audio-loop helper, not the rear composite-video RCA jack.

## The converted Apple keyboard

The vintage Apple keyboard has been converted to Bluetooth LE using Matt Chesters' keyboard-conversion project. It has its own battery, USB-C charging port, nice!view display and power switch near the upper-right edge. It communicates separately from the mouse receiver. Appendix A and the credits identify the conversion and its source.

## The matching mouse

The matching mouse combines Bambu wireless mouse-kit electronics with a printed Retro Wireless Computer Mouse shell. The scroll wheel and axle were replaced after the original wheel broke during modification. A USB receiver provides the connection to the computer.

## Bose speakers and stands

The Bose speakers are an optional extra that connect to the rear audio output socket. The PiForma's internal speaker works without them. Place them on stable supports with cables clear of the computer and its stand.

## The bag and its fitted interior

The carrying bag is part of the build. It has a corrugated-plastic liner, a foam bottom, green velvet, alignment grooves, and a wireless charging transmitter concealed under the lining. Its cable reaches the back pocket.

The foam locates the computer as well as cushioning it. The packing sequence in Chapter 1 protects the screen and puts the keyboard in its intended side position. A loose fit and a correctly aligned charging position are not interchangeable.

CHAPTER 3

# How the Macintosh PiForma Works

The PiForma wears a familiar face, but Raspberry Pi OS does the work underneath. This chapter explains the everyday business of pointing, typing, opening programs and keeping what you make. You need not learn the wiring to save a file.

## Finding your way around the desktop

The desktop is the working screen behind your application windows. PiForma Panel supplies the menu bar at the top; AtEase is the folder-like launcher for programs. The Control Strip provides another way to open or return to programs.

The desktop picture in Chapter 1 shows these parts together. The launcher has At Ease Items and sites tabs; Clippy occupies the lower-right corner. The clock appears at the upper right of the system bar.

Choose a menu by clicking its name, then click the command you want. An unavailable command may be dimmed. Read a dialog before answering: it may ask where to save a file, whether to discard changes, or whether to stop the computer.

The Macintosh appearance does not make every classic Macintosh command applicable. Use the commands shown by the running program. Its menus and the desktop's top menu bar can have different jobs.

## Using the mouse

Move the mouse across its pad to move the pointer. If you reach the edge, lift the mouse, set it down nearer the center and continue. The mouse need not point at the screen; the little eye underneath follows the surface below it.

Click means press and release the left button once. A click selects an item or operates a button. Double-click means two quick clicks without moving the pointer between them; many file views use it to open an item. AtEase program buttons need only one click. Repeated clicking while a program starts can produce more windows than you intended.

To drag, point at the object, hold the left button, move the mouse, then release. Drag a window by its title bar, rather than by its contents. Dragging a file may move or copy it, so practice with windows before rearranging your work.

Roll the wheel to scroll through a longer page or list. Point over the area you want to scroll. Scroll bars provide another way to reach material outside a window's visible area.

The right button may show a menu of actions for the item under the pointer. Its behavior depends on the program. This is the mysterious “second left click” in the illustration; it is a real right button, despite our best efforts. The Sync button is unnecessary for the normal one-receiver setup.

## Using the keyboard

Switch the keyboard on at its upper-right edge. It uses Bluetooth, independently of the mouse's USB receiver. First click in the window or text field where you want to type. The insertion point marks where new text will go; moving the mouse pointer alone does not move it.

Use the letter, number and punctuation keys to enter text. Hold Shift while typing a letter for a capital or a key's shifted symbol. Return commonly ends a paragraph in a document; in a dialog it may accept the current choice, so read the dialog first. Use the arrow keys to move through text where the program supports them.

To replace text, select it by dragging across it, then type the replacement. Deleting selected text removes the whole selection. If you remove something by mistake, look for Undo in that application's Edit menu before doing more work.

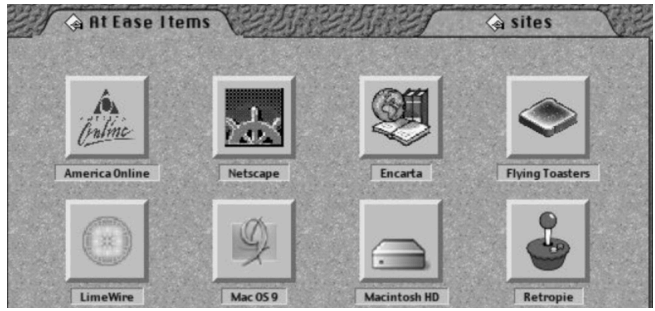
Use an application's own Cut, Copy and Paste commands when available. Copy leaves the original selection in place; Cut removes it; Paste inserts the copied or cut material at the insertion point. Check the selection before cutting, and the insertion point before pasting.

The vintage key legends do not establish a modern shortcut map. Follow commands displayed by the program instead of assuming that an Apple-labeled key performs a particular action. Pairing and special key combinations are not covered here. If ordinary typing fails, use the keyboard checks in Appendix B.

## Opening programs with AtEase

**AtEase**

AtEase has two folder tabs. At Ease Items holds the programs; sites holds a selection of websites the builder finds interesting, a collection still being tidied. Each tab shows only what fits in its window; there is no second page yet. Click a tab to switch between them, then click a button once. Allow time for the program to open. A launcher button starts a program; it is not the document you may later create inside it.




AtEase folder tabs and program buttons.

**Show Desktop**

If application windows cover the launcher, choose View → Show Desktop from PiForma Panel. This reveals the desktop without quitting the programs. You can then select another AtEase button. If the launcher is absent rather than covered, see Appendix B.




View → Show Desktop returns you to the launcher without quitting the open applications.

## Working with application windows

Several programs can be open at once. Click a window to bring it forward and direct your typing to it. Before typing a command or entering text, check that the intended window is active. An overlapping window is still open even when you can see only an edge of it.

Where a normal window has a title bar, drag it to a more useful position. Where a resize border or corner is provided, drag that border or corner to change the window's size. Some programs fix their window size or use the whole screen; those controls will not always be present.

Use a program's Quit or Exit command to leave it. A window close control closes that window, which may leave other windows from the same program running. A save question gives you a chance to keep changes; Cancel returns you to the work so you can decide.

Showing the desktop hides windows from view without saving or quitting them. Return to an open window by clicking its visible part or the appropriate Control Strip pane. Give the strip a moment to detect a newly opened program. Returning to a running program is different from deliberately opening another copy of it.

### Window controls

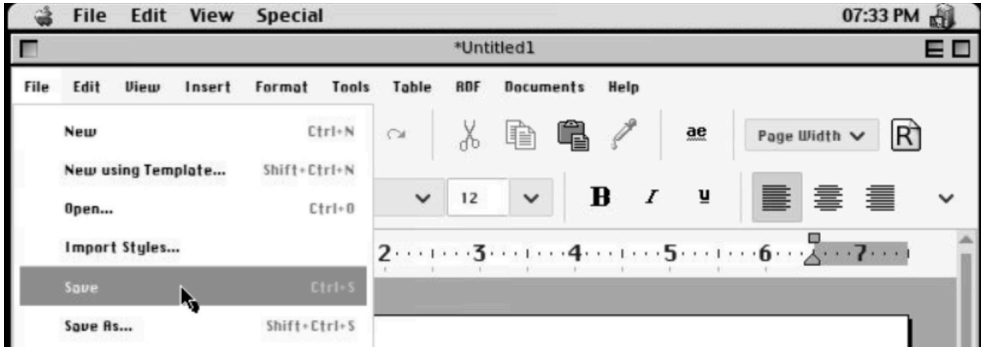
The themed document window pictured below has a close control at the upper left. At the upper right, the far-right control maximizes the window; the control immediately to its left minimizes it to the Dock, or Control Strip. Minimizing puts the window out of the way while leaving it open. Use the program's Control Strip pane to return to it.

Use the controls a window actually provides. Some windows omit controls or draw them differently. Clippy's chat, for example, has its close button at the upper right. Closing a window is not the same as minimizing it; save before closing work you want to keep.

### Two menu bars, two different jobs

PiForma OS uses a persistent system menu bar at the top of the desktop. Its system commands do not change to match the active application. This differs from the classic Macintosh global application menu bar, whose contents follow the active program.

Individual programs can provide application-local menu bars inside their own windows. For document work, use that application's own File → Save, Edit → Copy, Edit → Paste, and other available commands. The top system File menu and a document window's File menu have different jobs. Be sure you are opening the one you mean.

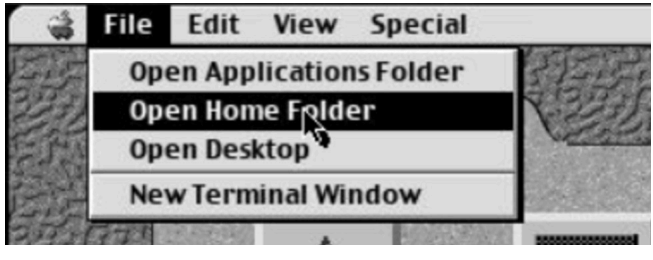



The persistent system menu bar remains above the document window. Use the application's own File menu for Save and its other document commands. The themed window's close control is at upper left; minimize and maximize are at upper right.

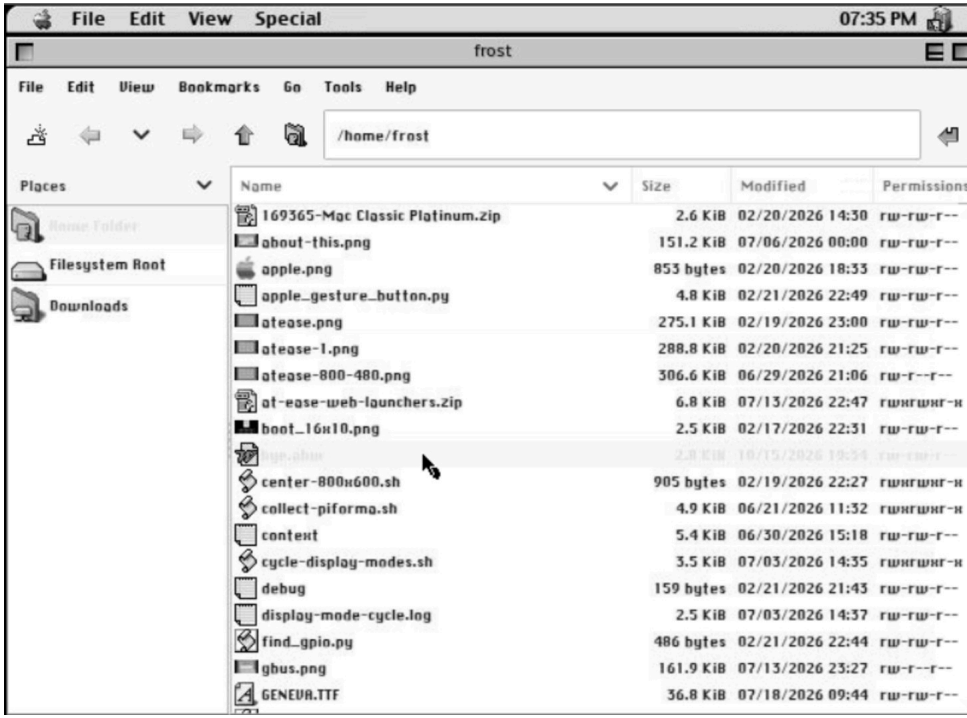
## Files and folders

A file holds something the computer can keep: a document, picture, sound or other information. A folder groups files and other folders. A program is the tool you use; the document is the work you keep. Closing the tool does not substitute for saving the work.

PiForma Panel's File → Open Home Folder opens your home folder in a file window. Browse by opening the folders shown there. Many file views open a folder or file with a double-click; use the view's Open command if its click setting differs. Opening a document starts the associated program when an association is available. An unfamiliar file type may instead ask which program to use.




Use the system File menu's Open Home Folder command to reach your files.




■ The home-folder window lists files and folders and has its own menu bar. The contents shown belong to this captured session; your list may differ.

Keep related work in a folder you can recognize. Give documents meaningful names and note which folder contains them. Before moving, renaming or deleting a file, close it in its program and make sure you know which item is selected. Leave unfamiliar system and application files alone.

Moving a file changes where you find it; copying leaves another copy. Neither should be confused with saving edits in an open document. The PiForma has no backup arrangement for its microSD card, and management prefers not to discuss it. The build is documented well enough to rebuild from scratch if the card ever fails, but your own files would not survive that. Copy any work you value to a USB drive or another computer. Two copies on the same card do not protect against failure of that card.

## Saving and reopening your work

Programs do not all save in the same way. Use this sequence in a document application that offers Save and Open commands; a web page or game may use a different system.

1. **Choose Save from the application's own menu.** On the first save, it will usually ask for a name and location.
2. **Select a folder you can find again, enter a useful filename, and confirm with the dialog's Save control.** Keep the format or filename extension the program supplies unless you understand the alternative.
3. **If warned that a file already exists, check its name and folder.** Replacing it may overwrite earlier work. Cancel and choose a different name if you want both versions.
4. **After further changes, use Save again.** Where offered, Save As lets you choose another name or location; read the dialog to see which copy you will be editing afterward.
5. **To return later, use the application's Open command and find the saved file, or open the file from its folder.** Check that the expected work appears.

Save periodically and before quitting or shutting down. A failed save, an error message or a dialog still waiting for an answer means you should check the result before closing. Where practical, reopen an important file to verify it. Do not assume an automatic save exists merely because no one has asked you to save yet.

## The menu bar and Control Strip

Apple menu → About This PiForma opens system information. View → Show Desktop reveals the desktop. Special → Shut Down requests an operating-system shutdown; Special → Sleep Display only blanks or sleeps the display. A sleeping screen is not a stopped computer.

The Control Strip sits at the bottom-left in the documented desktop arrangement. Click a configured program's pane to launch it when closed or return to its detected windows when open. Some running programs appear as temporary panes. Use the ordinary pane action; window-list details can differ between versions.




The Control Strip at lower left, with a window popup visible above it. The rest of the desktop is omitted so the strip can be read clearly.

## Apple-logo shortcuts

The Apple logo on the front is a touch control. Besides the six-second hold described under "Leaving a full-screen program," short taps open programs directly. Leave about half a second between taps.

| Apple logo | Opens |
|---|---|
| Tap once | The browser |
| Tap twice | Mac OS 9 |
| Tap three times | Winamp |
| Tap four times | RetroPie |
| Hold about six seconds | Closes the active application |

The RetroPie shortcut is the useful one. To play games without unpacking the keyboard and mouse, switch on, connect a Nintendo 64 controller, wait for the desktop and tap the logo four times.

## Controllers and games

The accessory pouch holds two Nintendo 64 controllers. The rear USB port and hub provide external USB connections. A controller with a different plug needs its matching supplied adapter; the shape of the controller does not establish its connection. Do not force a plug or substitute an unidentified adapter.



Two Nintendo 64 controllers and a racing game on the PiForma screen.

Open RetroPie/EmulationStation from AtEase to reach the game collection. Use the controls shown by the launcher to choose a system and game. Follow the game's own instructions for play. Controller assignments, player order and button meanings can differ between games; a button's label is not a universal promise about its action.

Before settling into a game, find out how that game pauses, saves and exits. Use the game's visible menu or established instructions. If a controller-configuration prompt appears, read it before proceeding; do not overwrite an existing setup merely to dismiss the prompt. If there are no usable prompts or known controls, ask the builder to identify the configured bindings.



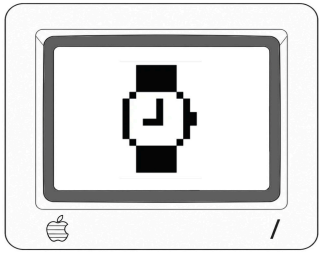

EmulationStation's Nintendo 64 game list. The bar along the bottom names the buttons for this screen, including A to launch the highlighted game and B to go back.

A game's own save and an emulator's save state are different facilities. Use only a save facility the running game or emulator actually provides, and check that it reports success. Closing a game from outside does not guarantee a saved game or state.

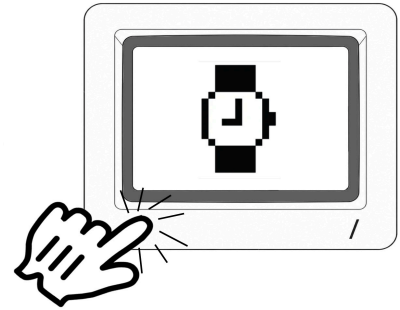
## Leaving a full-screen program

A full-screen game can cover the menu bar and window controls. First use its own Quit or Exit command when available. Save before leaving whenever the program allows it.

On the current PiForma, holding the Apple-logo touch control for about six seconds closes only the active application. Use this confirmed action when you need to leave a full-screen application without an accessible close control. It does not close everything or replace normal shutdown. Unsaved work may be lost.



An application that has stopped responding, showing only the busy watch.



Touch and hold the Apple logo for about six seconds to close the active application.

The same hold is how you quit a game. The Nintendo 64 controllers use every button, so none is set aside for quitting. Hold the Apple logo for about six seconds: the game closes and EmulationStation returns. To leave EmulationStation too, hold the logo for another six seconds; EmulationStation closes and the desktop returns.

Some emulators capture the mouse so the pointer stays inside their display. Use that emulator's documented release control if known. This guide supplies no unverified keyboard shortcut for releasing it. If you cannot regain control, follow the frozen-application checks in Appendix B.

## Diversions

The project includes a browser and a collection of themed diversions, with Clippy, Weather Channel, Flying Toasters and music-player experiments among them. Use the labels in your installed launcher. A familiar old name may open a modern program or themed web application rather than the original software. The costume department has been busy.

The browser is Chromium, dressed as Netscape Navigator; its AtEase button carries the Netscape name. Browser links and pages use their own controls. Enter an address in the browser's address field when you want a particular site; use the page's links to follow its contents. Online material needs a working network connection. Locally supplied diversions and documents may remain available without one.

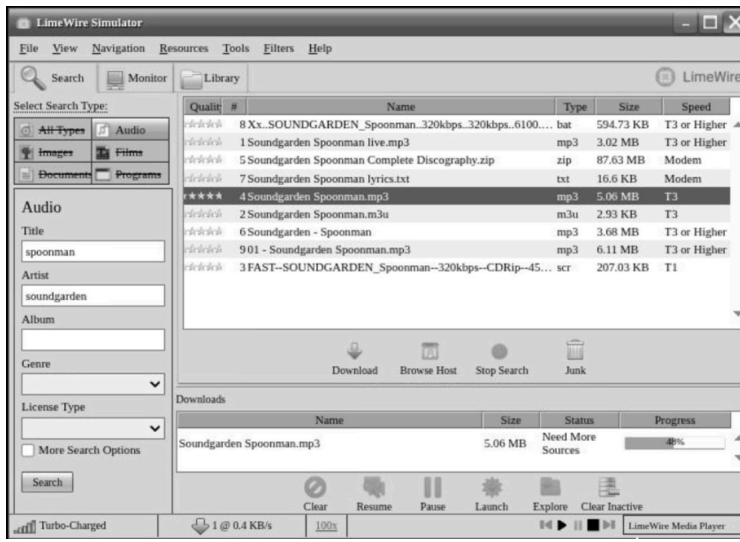
A few of the diversions are shown here: Winamp and its playlist, LimeWire Simulator, the Weather Channel and the Flying Toasters.

Winamp is the QMMP music player wearing a Winamp skin. It plays the songs in the Music folder in your home folder; to add more, copy them into that folder. The Weather Channel estimates your location from your internet connection and shows the nearest weather station. The Flying Toasters are a screen saver you start yourself. Because the animation works the computer hard, they never start on their own when the computer is idle. To stop them, hold the Apple logo for about six seconds.

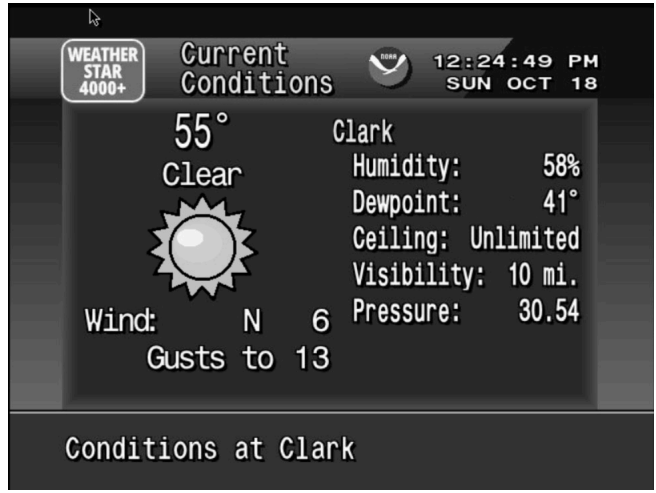
The other AtEase buttons are mostly what they seem. America Online opens an America Online simulator, and Encarta opens Kiwix, a reader for offline reference articles, at least as of this writing. Macintosh HD opens a file window. Mac OS 9 runs the real Mac OS 9 inside the SheepShaver emulator; it fills the screen, so leave it with the six-second hold. Terminal opens a command line, which is for the builder.



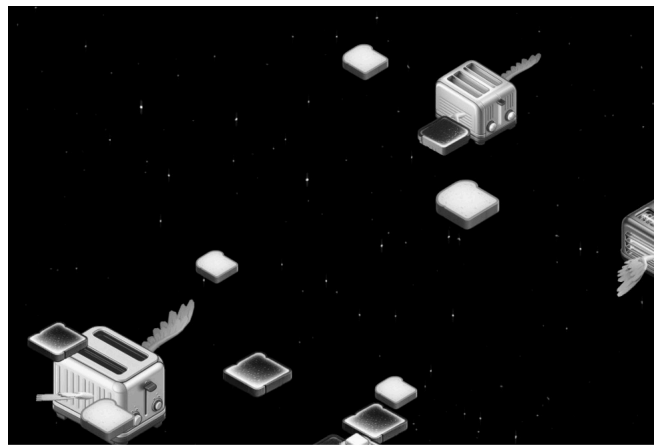

Winamp and its playlist window open over the AtEase launcher.




LimeWire Simulator, a period-styled take on the old file-sharing program.



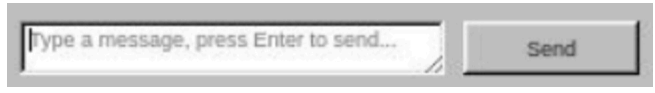

The Weather Channel diversion showing current conditions in WeatherStar 4000+ style.



The Flying Toasters, airborne.

## Clippy

Clippy is the assistant at the lower-right corner of the desktop. In the Chat with Clippy window, click the message field, type your question, then press Enter or click Send. The field itself reminds you how to send. Read the reply in the conversation area; use its scroll bar to revisit a longer conversation.




Enter a question in the message field, then press Enter or click Send.

Clippy can offer suggestions, but check them against the PiForma's actual setup before changing settings or installing anything. He is enthusiastic. We have given him his own corner of the screen, which has only encouraged him.




An example Chat with Clippy conversation. Its suggestions should be checked against the actual PiForma setup.

## About This PiForma and the project guide

About This PiForma displays computer and operating-system information. Open it from the Apple menu. It is useful when describing the machine to the builder, but it is not the same program as Macintosh PiForma Docs Viewer.

Docs Viewer presents the project's hardware, software, maintenance and other documents. Its Update Stack action checks for updated documentation. If an update fails, the viewer keeps its previous cached material; without a cache it can use its bundled offline snapshot. An offline document is useful, but its age still matters when comparing it with a changed machine.

The installed launcher calls this application PiForma Guide, and its window is titled Macintosh PiForma Guided Tour. Click PiForma Guide in AtEase, then choose a stack such as Meet PiForma, Hardware or Software to begin reading. It is the built-in guide to the computer and its functions. The bottom line reports documentation status and identifies the snapshot; it is worth checking when the machine and a document seem to disagree.




PiForma Guide opens Macintosh PiForma Guided Tour. Choose a documentation stack; the bottom line shows documentation status and the snapshot information.

## Wi-Fi and Bluetooth

Wi-Fi connects the computer to a wireless network. Bluetooth connects devices such as the keyboard. The mouse's USB receiver is a third, separate connection. A working keyboard does not prove that a website can be reached, and a lost internet connection does not mean the mouse needs pairing.

Use only a network you are authorized to join, with its correct name and password. Network menus depend on the installed desktop configuration; this guide does not specify an unconfirmed menu path or pairing sequence. For a failed connection, use Appendix B before changing saved settings.

## Ending a session

Save in each application, leave games and programs using their own exit commands, then choose Special → Shut Down. Let the operating system finish before switching off physical power, as described in Chapter 1. A menu disappearing or the display going dark is not by itself proof of completion. If a shutdown error appears, keep its wording for the builder.

CHAPTER 4

# Caring for Your Macintosh PiForma

The PiForma was built to be used and carried, with a little care. Its case has already had a long life; the electronics inside would prefer a reasonably quiet one.

## Handling the computer and display

Use the carrying handle as described in Chapter 1 and watch the keyboard as the computer comes out of the bag. Avoid impacts, pressure on the display, and pulling on connected cables.

The display connections need gentle handling. Do not try to cure a flicker by tapping the case or moving internal parts while powered. Record the symptom and let the builder investigate.

## Keyboard and mouse care

Keep crumbs, liquids, and loose items out of the keys. Pack the keyboard with its keys toward the bag wall, thick edge rear, and thin edge front, with Caps Lock up. Turn the mouse off for prolonged storage or transport if desired, and keep its receiver accounted for.

## Charging and storage

The owner prefers to keep the main battery above roughly 25–50% most of the time rather than repeatedly running it empty. This is a personal usage preference, not a manufacturer storage-charge specification.

Use the intended external supply. A full-looking battery may still need the plunger or a brief power connection before startup, as Chapter 1 describes. Bag charging requires the foam alignment. The battery display can indicate charging from the wireless receiver, but is hidden while the computer is packed in the bag. A USB power monitor can show power drawn by the bag charger; it does not directly measure what reaches the battery. Chapter 1 explains this limitation.

## Stand, bag and transport

Put the folding stand on a solid flat table, open it securely, and center the PiForma. The stand may permit some side-to-side motion, so keep it away from bumps and do not use it on an unstable surface.

Pack in the exact order in Chapter 1. Seat the computer in the bottom foam indents, orient the keyboard correctly, and replace the protective board before the pad, guide, mouse/stand, and pouch. The bag’s padding helps; it does not justify dropping it.

## Bose speakers and stands

The speakers travel separately from the bag. Move the speakers and stands carefully and keep their cables from pulling on the computer. Do not assume the stands can be carried by the speakers or folded while attached.

## Cleaning, ventilation and unusual behavior

Switch off and disconnect charging power before ordinary exterior cleaning. Use a soft dry cloth gently on the case; keep liquids and solvents away from openings and internal parts. Confirm the LCD's cleaning instructions before using a screen-cleaning product.

Leave vents clear and watch for unusual heat, smell, swelling, damaged wiring, or electrical behavior that is out of character. Stop using the machine in that condition. The build's heat history is a reason to investigate, not to ignore a problem because the desktop still appears.

## Service and support

There isn't any. Sorry.

Only one Macintosh PiForma was ever made, and it was never offered for sale. There is no service department, warranty desk, or helpful person in a tie waiting by the phone. If it needs repair, it will probably return to the person who built it, assuming he is in the mood.

That does not make unusual heat or electrical damage a joke. Stop using an unsafe-looking machine and speak to the builder rather than continuing to experiment. Treat it kindly and perhaps neither of us will have to find out how extensive the service department's paperwork is.

# Appendix A: Technical Information

This appendix describes the PiForma's hardware, connections and software. Use the project documentation for maintenance and recovery instructions.

## Hardware inventory and identification

| | |
|---|---|
| **Donor:** | Apple IIc Monitor A2M4090, February 1985. |
| **Computer:** | Raspberry Pi 4 Model B, 4 GB RAM. |
| **Storage:** | 256 GB microSD. |
| **Display:** | 9-inch LCD kit, working mode 800 × 480; about 640 × 480 shows inside the bezel, a little more across. Software is sized for 640 × 480. Brightness and color are fixed. |
| **Size:** | the original monitor's exterior, unchanged. |
| **Weight:** | 8 lb 13 oz (4.0 kg), close to the original monitor. |
| **Battery runtime:** | typically about six hours, or about four hours of emulated games. |
| **Main battery:** | Anker Prime 26K 300W. |
| **Internal charger:** | 100W USB-C adapter. |
| **Keyboard:** | Apple A9M0330 ADB keyboard, BLE conversion, ZMK, USB-C charging, nice!view. |
| **Mouse:** | Bambu kit electronics in a printed shell, USB receiver. |
| **Startup sound:** | YS-M3, CK1028 selector, separate salvaged speaker. |
| **Normal sound:** | internal USB speaker, or the rear audio output socket for speakers or headphones. |

## Power architecture

Power path: rear AC inlet → internal charger → main battery → original power switch → custom 5V JST rail → Pi, LCD controller, startup-chime module, and touch sensor. The USB speaker is supplied through the Pi. The switch does not disconnect the AC charging path. The rail has no separate installed fuse protection.

## Connections and GPIO

| | |
|---|---|
| **Apple-logo sensor:** | TTP223 on BCM GPIO23, active-low. |
| **EC11 volume encoder:** | BCM GPIO17/27; push input GPIO22 exists in software but is not physically usable on the installed knob. |
| **Display source:** | physical switch to LCD controller source control, not a GPIO relay selector in this build. |
| **Composite RCA input:** | video only. |
| **Rear audio socket:** | output to external speakers or headphones. |
| **External audio:** | USB capture source and audio-loop helper. |
| **External USB access:** | one rear port; do not infer exterior access from the Pi's board connector count. |

## Software, services and recovery

The software includes Raspberry Pi OS, X11/LXDE, Openbox, LightDM, PCManFM, PipeWire, and WirePlumber. System services include apple-gesture-button, volume-knob, atease, and flyingtoasters. User services include piforma-panel, controlstrip-simulator, and the optional usb-audio-loop.

Use the canonical repository's software page, service inventory, configuration archive and installation guide for recovery. Archived helpers describe an earlier installation; do not substitute them for deployed scripts without checking the running configuration.

## Keyboard conversion and sources

The build credits Matt Chesters' zmk-apple-desktop-bus-keyboard conversion. Special keys follow the deployed keymap.

The keyboard conversion uses a Seeed XIAO nRF52840, MCP23017 I/O expander, custom board, small battery and nice! view display.

# Appendix B: When Something Goes Wrong

Start with what you can see and hear. Make one check at a time, then observe the result. Note any message before dismissing it. Stop using the PiForma if there is unusual heat, swelling, a hot smell or damaged power wiring; those symptoms are a reason to speak to the builder, not try another restart.

## No green light, chime or picture

Check the right-side physical power control. Press the rear battery plunger and read the battery percentage through the top slits. Try the intended external power connection, then retry startup. A battery that previously showed full has sometimes needed a brief power connection; Chapter 1 describes this observation without promising a recovery time. If there is still no response, stop short of internal adjustments and consult the builder.

## Battery does not appear to charge

Check the intended cable and external power connection without opening the computer. For the bag arrangement, check the rear/bottom seating in the foam indents and the charging cable in the back pocket. The battery display can indicate charging from the wireless receiver, but you cannot see it while the computer is packed in the bag. A USB power monitor on the bag charger's supply can show power draw, but does not directly measure charging at the battery. Without that additional evidence, whether the bag is charging remains uncertain. Do not substitute an unidentified supply.

## Green light or chime, but no desktop

Allow approximately one to two minutes for startup, with some variation. Check that the rear display source is set for HDMI. A chime does not prove that the computer has finished booting. If the startup picture persists, note where it stops. If the image is blank, flickers or distorts, do not tap the case or adjust internal cables while powered. Ask the builder to investigate a persistent display problem.

## Mouse pointer does not move

Check the underside power switch and put the mouse on its pad. Keep the optical tracking window facing the pad. Check that the mouse's USB receiver is present and seated where accessible without dismantling the computer; check any accessible hub connection. The Sync button is not part of normal setup. If the pointer works elsewhere but stays inside one emulator, it may be captured; see Chapter 3 rather than assuming the mouse has failed.

## Keyboard does not type

If letters come out as capitals or the number keys type symbols, Caps Lock is down: press it again so it sits level with the other keys. Otherwise, check the switch near the upper-right edge and click in the intended text field. Try another available text field. The keyboard uses Bluetooth; a working mouse does not prove that connection. If typing fails everywhere, check keyboard power, charge and connection with the builder. Use the configured pairing procedure, not guessed key combinations.

## No sound, or no startup chime

For normal program sound, turn the side volume control clockwise a little and check that the program has something playing and its own sound is enabled. The installed side knob has no usable push-to-mute action. Keep the initial volume modest. If sound has been switched to the rear audio socket, it plays only through whatever is plugged in there, not the internal speaker.

For a missing startup chime, remember that some rear selector positions are silent. Follow Chapter 1's powered-on selector check without expecting immediate playback at every position. The chime and normal program sound use separate hardware; hearing one does not prove that the other is working. Describe which sound is missing before asking for help.

## A website or network service will not open

Try another known page or service. Check the visible connection status and intended network. If another device on that network also fails, the problem may be outside the PiForma. Bluetooth and the mouse receiver do not establish Wi-Fi access. Keep error wording; ask for the installed reconnect procedure if needed.

## A program will not respond

Wait briefly, then look for a dialog waiting behind the main window. Check whether the pointer and other applications respond. Save unaffected work if you can and try the troubled program's own Quit or close control.

If necessary, the owner-confirmed Apple-logo hold of about six seconds closes only the active application. Check which application is active first. Unsaved work may be lost; this does not guarantee a game save. Use View → Show Desktop to uncover the launcher when the menu bar is accessible.

If the entire desktop remains unresponsive, use Special → Shut Down if you can reach it. Abrupt physical power-off is a last resort and may lose work or damage files. A sleeping screen is not completed shutdown. Note what was running and what stopped responding.

If the launcher or menu bar does not return after leaving a full-screen program, save accessible work and consult the builder. Do not try longer logo holds.

# Glossary

**Apple Desktop Bus (ADB):** The connection system used by the original vintage keyboard before its Bluetooth conversion.

**AtEase:** The PiForma's folder-like launcher for opening configured programs.

**Bluetooth LE:** Bluetooth Low Energy, the wireless connection used by the converted keyboard.

**Chime selector:** The rear dial that chooses a startup sound; some settings are silent.

**Clippy:** The assistant at the lower-right corner of the desktop. Type a question in its Chat with Clippy window; check its suggestions against the actual setup.

**Composite video:** The video signal accepted by the rear RCA input, used to show an external source on the PiForma screen. The jack carries video only, not audio.

**Control Strip:** The small program-launching and window-focus strip supplied by ControlStrip Simulator.

**Desktop:** The working screen with its background, launchers, menu bar, and application windows.

**Docs Viewer:** Macintosh PiForma Docs Viewer, the program for reading the project's hardware, software and maintenance documents, with an offline snapshot. It is separate from About This PiForma.

**EmulationStation:** The game-list menu of RetroPie, opened from AtEase. It is a separate application from the games it starts.

**Emulator:** A program that imitates another computer or games system so compatible software can run.

**Full-screen application:** A program occupying the display, often covering the normal desktop menus.

**GPIO:** General-purpose input/output pins on the Raspberry Pi, used for controls in this build. Appendix A uses BCM pin numbering.

**HDMI:** The digital video connection that carries the Raspberry Pi's picture to the PiForma screen. Select HDMI on the rear source selector to see the Pi.

**microSD:** The small storage card containing the PiForma's operating system and saved files.

**PiForma Panel:** The program that supplies the menu bar across the top of the screen, including the Special menu with Shut Down and Sleep Display.

**Qi charging:** Wireless charging using an aligned transmitter and receiver.

**Raspberry Pi:** The small computer board inside the PiForma case.

**Raspberry Pi OS:** The operating system documented beneath the PiForma's Macintosh-inspired desktop.

**RetroPie:** The games/emulation environment, opened through EmulationStation in this setup.

**Save state:** An emulator's snapshot of a game in progress. It is a different facility from the game's own save.

**Screen saver:** An animation that fills the screen. On the PiForma, the Flying Toasters are a screen saver you start yourself; nothing starts on its own when the computer is idle. Hold the Apple logo for about six seconds to stop it.

**USB hub:** An accessory that provides several USB connections from one computer port.

**Wi-Fi:** The wireless network connection, managed by the operating system's network tools. Bluetooth and the mouse receiver do not provide it.

# Credits and acknowledgments

This build uses many good ideas by other people, adapted to a computer-shaped joke that became a useful project. The Apple manuals supplied visual and organizational references, not PiForma specifications or approval. Apple had no part in the PiForma; the story in the welcome is a tall tale. Their text has not been copied into this guide.

- Canonical build records: Macintosh-PiForma-docs.
- Keyboard conversion credited in those records: Matt Chesters' ZMK ADB keyboard project.
- Mouse shell credited in the hardware record: Retro Wireless Computer Mouse.
- Replacement wheel/axle credited there: Custom scroll wheel and axle.
- Companion applications: PiForma Panel, AtEase Simulator, ControlStrip Simulator, About This PiForma, and Docs Viewer.
- The canonical related-projects page credits the Clippy fork to Felix Rieseberg's upstream project and links the Flying Toasters application.

## PiForma Limited Warranty

The builder warrants that this PiForma was a working computer when it left the workbench. That is the whole of the warranty.

There is no repair turnaround time and no promise that anything that breaks will ever be fixed. Repairs, if any, happen when the builder has a free weekend and the right part. Opening the Forbidden Latch voids this warranty, which costs you nothing, because there was never much of one to void.

In keeping with the company's priorities of the era, this warranty is offered to consumers only. Enterprise customers should consult their enterprise agreement, of which there is none.

## Radio and television interference

The PiForma contains Wi-Fi and Bluetooth radios, a wireless charging receiver and a Raspberry Pi, all of which give off radio-frequency energy. Unlike the computers this guide imitates, the assembled PiForma has not been tested against the FCC limits for a Class B computing device.

If it interferes with radio or television reception, which you can find out by switching it off and on, move it farther from the receiver. Alternatively, switch off the television and play a tape through the PiForma's rear composite input instead.

## Tell us what you think

Apple's manuals ended with a reply card asking what you thought. This one does too, without the postage. If you have looked through the PiForma or this guide and have a thought, a question or a correction, tell the builder in person or open an issue on the project's documentation repository, listed in the credits above.

Compliments are forwarded to the CEO. Complaints are forwarded to the rogue engineer.